

# Attack–Fault Ambiguity Against a Strategic Adversary: Identifiability, Harm Ceilings, and Verification

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*Abstract—*

*Draft pending.* Claim: against a committed belief-based operator, an  $\varepsilon$ -stealthy attacker faces a detection–harm frontier; exact stealth is generically impossible under state feedback, and verification shifts the frontier against the attacker.

## I. INTRODUCTION

*Planned.* Papers 1–3 treat the attacker as fixed; a leader–follower formulation answers whether the framework survives an adversary who knows the rule.

## II. RELATED WORK

*Planned.* Stealthy attacks in cyber-physical systems, security games, adversarial classification.

## III. MODEL

*Planned.* Stackelberg game; attacker constrained to total-variation distance  $\varepsilon$  from the fault law. Code: `src/paper4_game`.

## IV. THE DETECTION–HARM FRONTIER

*Planned.* Frontier; harm ceiling as the  $\varepsilon \rightarrow 0$  limit (Theorem 1 output-level; zero generically under Theorem 2).

## V. NUMERICAL ILLUSTRATION

*Planned.*

## VI. DISCUSSION

*Planned.*

## VII. CONCLUSION

*Planned.*

## REFERENCES

- [1] F. Pasqualetti, F. Dörfler, and F. Bullo, “Attack detection and identification in cyber-physical systems,” *IEEE Transactions on Automatic Control*, vol. 58, no. 11, pp. 2715–2729, 2013.
- [2] T. Roy and S. Dey, “On distinguishability of anomalies as physical faults or actuation cyberattacks,” *ASME Letters in Dynamic Systems and Control*, vol. 4, no. 3, p. 031003, 2024, arXiv:2311.09342.